


Lecture # 10

Project Cost Management

ENGINEERING PROJECT MANAGEMENT

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Project Cost Management

- Project Cost Management includes the processes involved in planning, estimating, budgeting, financing, funding, managing, and controlling costs so that the project can be completed within the approved budget

Project Cost Management - Processes

- 7.1 Plan Cost Management—The process of defining how the project costs will be estimated, budgeted, managed, monitored, and controlled
- 7.2 Estimate Costs—The process of developing an approximation of the monetary resources needed to complete project work
- 7.3 Determine Budget—The process of aggregating the estimated costs of individual activities or work packages to establish an authorized cost baseline
- 7.4 Control Costs—The process of monitoring the status of the project to update the project costs and manage changes to the cost baseline

Project Cost Management

Table 1-4. Project Management Process Group and Knowledge Area Mapping

Knowledge Areas	Project Management Process Groups				
	Initiating Process Group	Planning Process Group	Executing Process Group	Monitoring and Controlling Process Group	Closing Process Group
4. Project Integration Management	4.1 Develop Project Charter	4.2 Develop Project Management Plan	4.3 Direct and Manage Project Work 4.4 Manage Project Knowledge	4.5 Monitor and Control Project Work 4.6 Perform Integrated Change Control	4.7 Close Project or Phase
5. Project Scope Management		5.1 Plan Scope Management 5.2 Collect Requirements 5.3 Define Scope 5.4 Create WBS		5.5 Validate Scope 5.6 Control Scope	
6. Project Schedule Management		6.1 Plan Schedule Management 6.2 Define Activities 6.3 Sequence Activities 6.4 Estimate Activity Durations 6.5 Develop Schedule		6.6 Control Schedule	
7. Project Cost Management		7.1 Plan Cost Management 7.2 Estimate Costs 7.3 Determine Budget		7.4 Control Costs	
8. Project Quality Management		8.1 Plan Quality Management	8.2 Manage Quality	8.3 Control Quality	

PLAN COST MANAGEMENT

- The key benefit of this process is that it provides guidance and direction on how the project costs will be managed throughout the project

PLAN COST MANAGEMENT

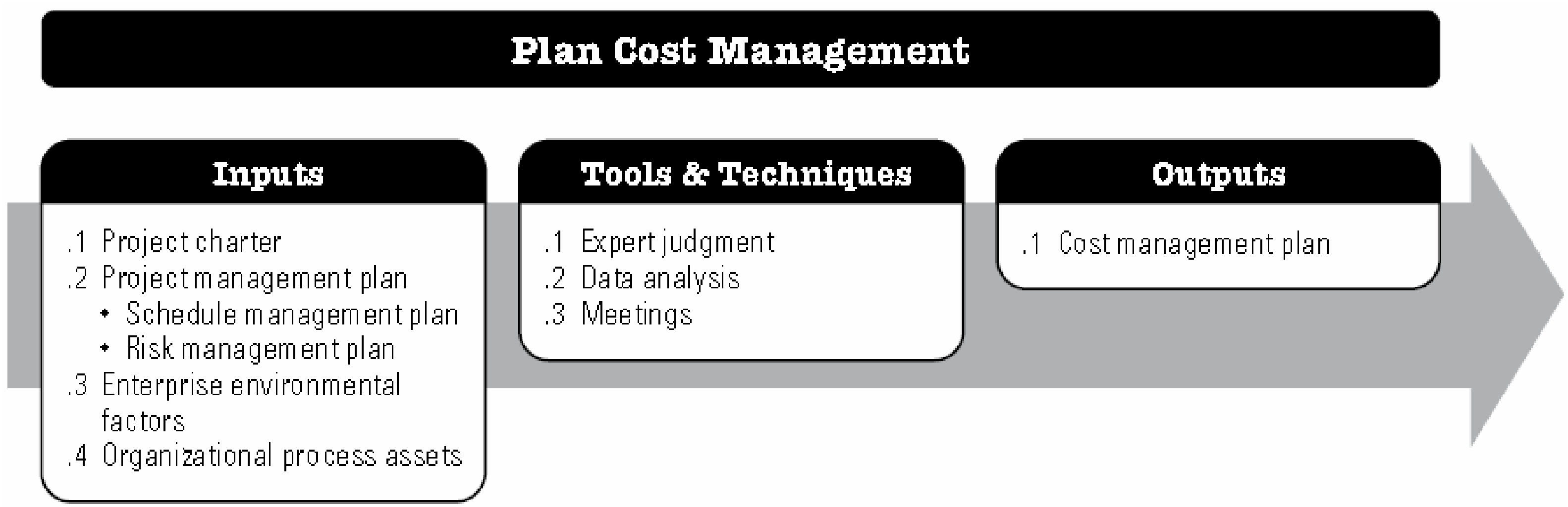


Figure 7-2. Plan Cost Management: Inputs, Tools & Techniques, and Outputs

Estimate Costs

- The key benefit of this process is that it determines the monetary resources required for the project

Estimate Costs

Estimate Costs

Inputs

.1 Project management plan
- Cost management plan
- Quality management plan
- Scope baseline

.2 Project documents
- Lessons learned register
- Project schedule
- Resources requirements
- Risk register

.3 Enterprise environmental factors

.4 Organizational process assets

Tools & Techniques

.1 Expert judgment

.2 Analogous estimating

.3 Parametric estimating

.4 Bottom-up estimating

.5 Three-point estimating

.6 Data analysis
- Alternatives analysis
- Reserve analysis
- Cost of quality

.7 Project management information system

.8 Decision making
- Voting

Outputs

.1 Cost estimates

.2 Basis of estimates

.3 Project documents updates
- Assumption log
- Lessons learned register
- Risk register

Figure 7-4. Estimate Costs: Inputs, Tools & Techniques, and Outputs

Estimate Costs

- The accuracy of a project estimate will increase as the project progresses through the project life cycle. For example,
  - A project in the initiation phase may have a **rough order of magnitude (ROM) estimate** (ROM) in the range of −25% to +75%.
  - In the middle of the project, budget estimates -- could narrow the range of accuracy to −10% to +25%.
  - Later in the project, as more information is known, **definitive estimates** could narrow the range of accuracy to −5% to +10%.
- Costs are estimated for all resources that will be charged to the project. This includes but is not limited to
  - labor, materials, equipment, services, and facilities, as well as
  - special categories such as an inflation allowance, cost of financing, or contingency costs

Estimate Costs – Tools and Techniques

(1) ANALOGOUS ESTIMATING:

- Analogous estimating is a technique for estimating the duration or cost of an activity or a project using historical data from a similar activity or project.
- Analogous duration estimating is frequently used to estimate project cost when there is a limited amount of detailed information about the project.
- Analogous estimating is generally less costly and less time-consuming than other techniques, but it is also less accurate.

(2) PARAMETRIC ESTIMATING:

- Parametric estimating is an estimating technique in which an algorithm is used to calculate cost or duration based on historical data and project parameters.
- This technique can produce higher levels of accuracy depending on the sophistication and underlying data built into the model.
- Parametric schedule estimates can be applied to a total project or to segments of a project, in conjunction with other estimating methods.

Estimate Costs – Tools and Techniques

3) THREE-POINT ESTIMATING:

- The accuracy of single-point cost estimates may be improved by considering estimation uncertainty and risk.
- Using three-point estimates helps define an approximate range for an activity's cost:
- **Most likely (c*M*).** The cost of the activity, based on realistic effort assessment for the required work and an predicted expenses.
- **Optimistic (c*O*).** The cost based on analysis of the best-case scenario for the activity.
- **Pessimistic (c*P*).** The cost based on analysis of the worst-case scenario for the activity.
- Depending on the assumed distribution of values within the range of the three estimates, the expected cost, *cE*, can be calculated. One commonly used formula is triangular distribution:
  - $cE = (cO + cM + cP) / 3$.
- Triangular distribution is used when there is insufficient historical data or when using judgmental data. Cost estimates based on three points with an assumed distribution provide an expected cost and clarify the range of uncertainty around the expected cost.

Estimate Costs – Tools and Techniques

(4) BOTTOM-UP ESTIMATING:

- Bottom-up estimating is a method of estimating project duration or cost by aggregating the estimates of the lower level components of the WBS

(5) DATA ANALYSIS - **Reserve analysis**

- Cost estimates may include contingency reserves (sometimes called contingency allowances) to account for cost uncertainty. Contingency reserves are the budget within the cost baseline that is allocated for identified risks.
  - **Contingency reserves** are often viewed as the part of the budget intended to address the **known unknowns** that can affect a project.
  - For example, rework for some project deliverables could be anticipated, while the amount of this rework is unknown.
  - Contingency reserves may be estimated to account for this unknown amount of rework.

Determine Budget

- The key benefit of this process is that it determines the cost baseline against which project performance can be monitored and controlled

Determine Budget

Determine Budget

Inputs

.1 Project management plan
- Cost management plan
- Resource management plan
- Scope baseline

.2 Project documents
- Basis of estimates
- Cost estimates
- Project schedule
- Risk register

.3 Business documents
- Business case
- Benefits management plan

.4 Agreements

.5 Enterprise environmental factors

.6 Organizational process assets

Tools & Techniques

.1 Expert judgment

.2 Cost aggregation

.3 Data analysis
- Reserve analysis

.4 Historical information review

.5 Funding limit reconciliation

.6 Financing

Outputs

.1 Cost baseline

.2 Project funding requirements

.3 Project documents updates
- Cost estimates
- Project schedule
- Risk register

Figure 7-6. Determine Budget: Inputs, Tools & Techniques, and Outputs

Output – cost baseline

- Cost estimates for the various project activities, along with any contingency reserves for these activities, are aggregated into their associated
  - work package costs. The work package cost estimates, along with any contingency reserves estimated for the work packages, are aggregated into
  - control accounts. The summation of the control accounts make up the **cost baseline**.

Control Costs

- The key benefit of this process is that the cost baseline is maintained throughout the project

Control Costs

Control Costs

Inputs

.1 Project management plan
- Cost management plan
- Cost baseline
- Performance measurement baseline

.2 Project documents
- Lessons learned register

.3 Project funding requirements
.4 Work performance data
.5 Organizational process assets

Tools & Techniques

.1 Expert judgment
.2 Data analysis
- Earned value analysis
- Variance analysis
- Trend analysis
- Reserve analysis

.3 To-complete performance index
.4 Project management information system

Outputs

.1 Work performance information
.2 Cost forecasts
.3 Change requests
.4 Project management plan updates
- Cost management plan
- Cost baseline
- Performance measurement baseline

.5 Project documents updates
- Assumption log
- Basis of estimates
- Cost estimates
- Lessons learned register
- Risk register

Figure 7-10. Control Costs: Inputs, Tools & Techniques, and Outputs

TOOLS AND TECHNIQUES – **EVA**

- **Earned value analysis (EVA).** Earned value analysis compares the performance measurement baseline to the actual schedule and cost performance.
- EVM develops and monitors three key dimensions for each work package and control account:
  - **Planned value** (PV)- is the authorized budget assigned to scheduled work. It is the authorized budget planned for the work to be accomplished for an activity or work breakdown structure (WBS) component, not including management reserve. The total of the PV is sometimes referred to as the performance measurement baseline (PMB). The **total planned value** for the project is also known as budget at completion (BAC).
  - **Earned value** (EV) is a measure of work performed expressed in terms of the budget authorized for that work. It is the budget associated with the authorized work that has been completed. The EV is often used to calculate the percent complete of a project.
  - **Actual cost** (AC) is the realized cost incurred for the work performed on an activity during a specific time period. It is the total cost incurred in accomplishing the work that the EV measured.

TOOLS AND TECHNIQUES – **EVA**

- Variance analysis, as used in EVM, is the explanation for cost (CV = EV – AC), schedule (SV = EV – PV), and **variance at completion** (VAC = BAC – EAC) variances.
  - **Schedule variance** (SV) is a measure of schedule performance expressed as the difference between the earned value and the planned value. It is the amount by which the project is ahead or behind the planned delivery date, at a given point in time. It is a measure of schedule performance on a project. Equation: **SV = EV – PV**
  - **Cost variance** (CV) is the amount of budget deficit or surplus at a given point in time, expressed as the difference between earned value and the actual cost. It is a measure of cost performance on a project. It is equal to the earned value (EV) minus the actual cost (AC).
  - The CV is particularly critical because it indicates the relationship of physical performance to the costs spent. Negative CV is often difficult for the project to recover. Equation: **CV = EV – AC**.

TOOLS AND TECHNIQUES – **EVA**

- Variance analysis, as used in EVM, is the explanation for cost (CV = EV – AC), schedule (SV = EV – PV), and **variance at completion** (VAC = BAC – EAC) variances.
  - **Schedule performance index** (SPI) is a measure of schedule efficiency expressed as the **ratio of earned value to planned value**. It measures how efficiently the project team is accomplishing the work.
  - An SPI value **less than 1.0** indicates less work was completed than was planned. An SPI **greater than 1.0** indicates that more work was completed than was planned. Since the SPI measures all project work, the performance on the critical path also needs to be analyzed to determine whether the project will finish ahead of or behind its planned finish date. The SPI is equal to the ratio of the EV to the PV. Equation: **SPI = EV/PV**.
  - **Cost performance index** (CPI) is a measure of the cost efficiency of budgeted resources, expressed as a **ratio of earned value to actual cost**.
  - A CPI value of **less than 1.0** indicates a cost overrun for work completed. A CPI value **greater than 1.0** indicates a cost underrun of performance to date. The CPI is equal to the ratio of the EV to the AC. Equation: **CPI = EV/AC**

TOOLS AND TECHNIQUES – **Trend Analysis**

- **Trend analysis** examines project performance **over time** to determine if performance is improving or deteriorating.
- Graphical analysis techniques are valuable for understanding performance to date and for comparison to future performance goals in the form of BAC versus estimate at completion (EAC) and completion dates.
- Examples of the trend analysis techniques include but are not limited to: *Charts, forecasting*

TOOLS AND TECHNIQUES – **Trend Analysis**

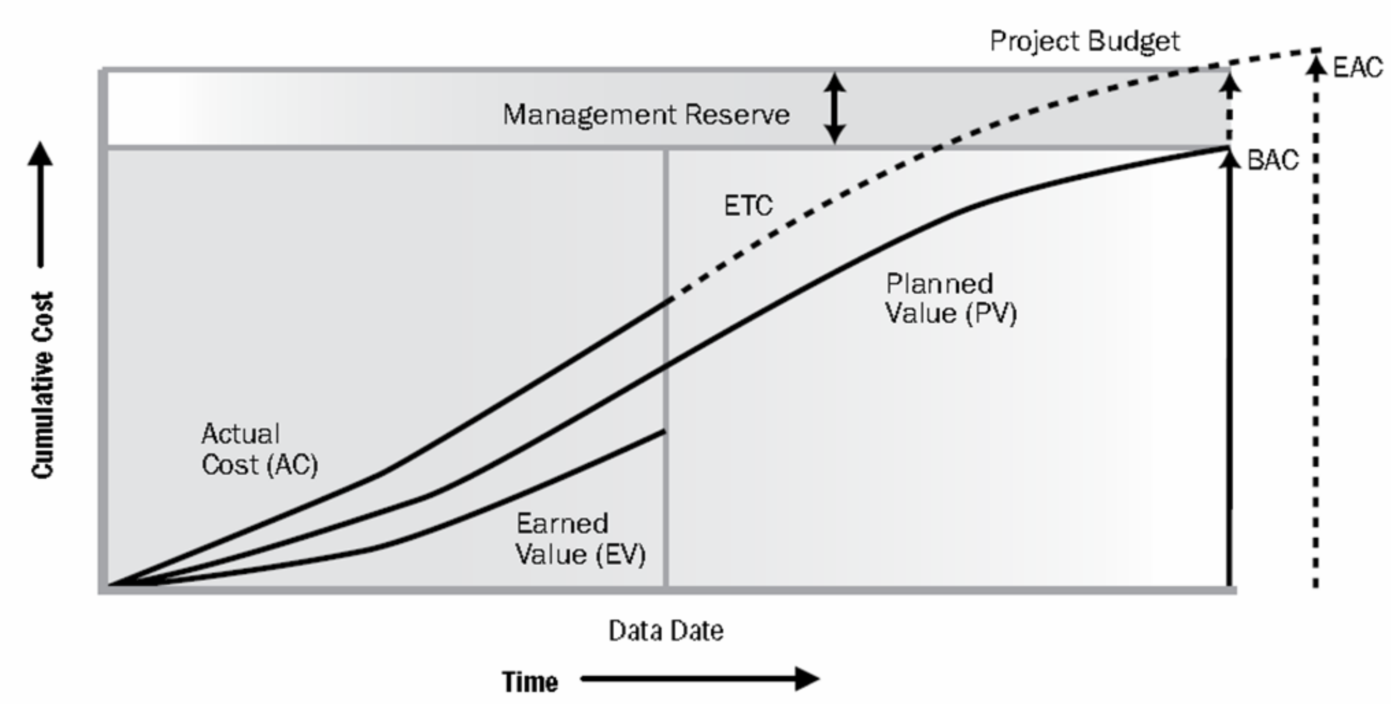


Figure 7-12. Earned Value, Planned Value, and Actual Costs

EVA formulas

- Schedule Variance, **SV** = EV – PV
- Cost Variance, **CV** = EV – AC
- Cost Performance Index, **CPI** = EV/AC
- Schedule Performance Index, **SPI** = EV/PV

BAC is estimated at the very beginning of the project; but, EAC is at any time during the project execution.

Estimate at Completion, EAC:

- 1. **EAC** = BAC/CPI
- 2. **EAC** = AC + (BAC – EV)
- 3. **EAC** = AC + (BAC – EV)/ (CPI x SPI)
- 4. **EAC** = AC + New Cost Estimation

Smaller TCPI – high probability of completion within budget; a "future performance"

To-Complete Performance Index, TCPI

- **TCPI** = (BAC – EV)/(BAC – AC) ---- (the CPI rate to comple… project within the **original budget)**

If an CPI is negative (=not good!); then we need the future cost performance, TCPI value to be greater than 1.

Estimate to Complete, ETC:

- **ETC** = EAC – AC ---- (money spent until the forecasted budget is reached, by the end of the project)

Variance at Completion , VAC:

- **VAC** = BAC – EAC ---- (amount of money spent under or over the budget, by the end of the project)

Tools and Techniques

Important Results:

Cost Variance (CV = EV – AC)

CV > 0 ➔ Good (Under budget)
CV = 0 ➔ Good (Just within the limits)
CV < 0 ➔ Bad (Over budget)

Schedule Variance (SV = EV – PV)

SV > 0 ➔ Good (Ahead of schedule)
SV = 0 ➔ Good (Just as planned)
SV < 0 ➔ Bad (Behind of schedule)

Cost Performance Index (CPI = EV / AC)

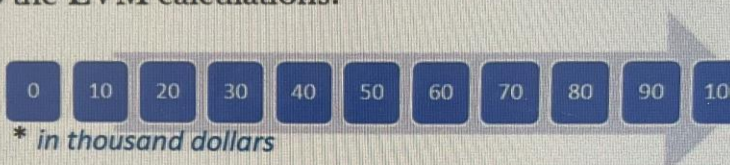
CPI>1 ➔Good (Under budget)
CPI=1 ➔Good (Just within the limits)
CPI<1 ➔Bad (Over budget)

Schedule Performance Index (SPI = EV / PV)

SPI>1 ➔Good (Ahead of schedule)
SPI=1 ➔Good (Just as planned)
SPI<1 ➔Bad (Behind of schedule)

Tools and Techniques

Example: Pipeco Incorporated has been awarded a piping project to install 1,000 meters of pipeline in 100 days with a cost of $100,000. Today is the 30th day of the project, and Pipeco has installed 400 meters of piping work up to now. Also, the report says $20,000 has been spent until now. Let's do the EVM calculations.



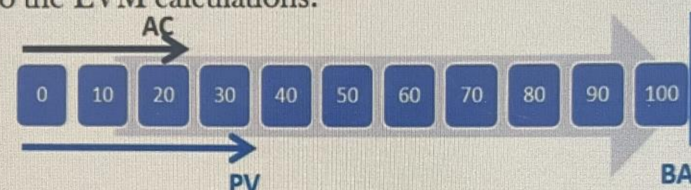
- Assume the company simply prepared the schedule to install 10 meters each day
- Pipeco will spend $1,000 each day

- Scope Baseline
- Schedule Baseline
- Cost Baseline

Performance Measurement Baseline

➔ If we spend $800 each day this is good, since it is less than $1,000.
➔ If we install 8.00 meters of pipeline each day, this is bad, since it is less than 10 meters.

Example: Pipeco Incorporated has been awarded a piping project to install 1,000 meters of pipeline in 100 days with a cost of $100,000. Today is the 30th day of the project, and Pipeco has installed 400 meters of piping work up to now. Also, the report says $20,000 has been spent until now. Let's do the EVM calculations.



- Assume the company simply prepared the schedule to install 10 meters each day
- Pipeco will spend $1,000 each day

$100,000	= Budget	➔ BAC
$30,000	= Planned Value	➔ PV
$20,000	= Actual Cost	➔ AC

How much money should we spend to install 400 meters of pipeline?

➔ 100,000 / 1,000 = $100/meters
➔ 400 x 100 = **$40,000**

Tools and Techniques

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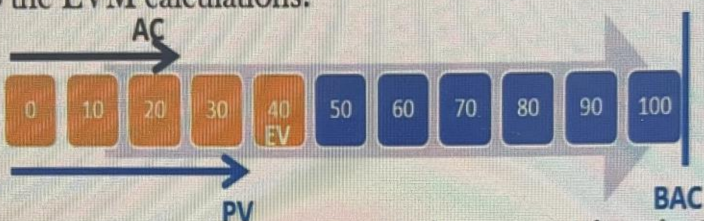
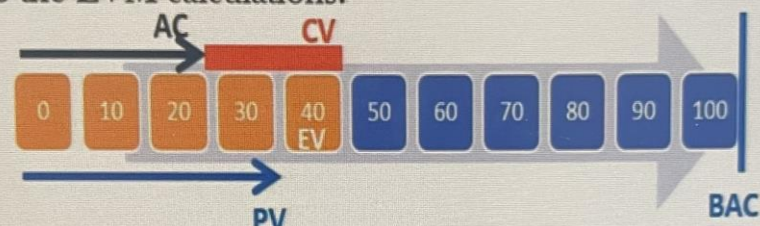


- Assume the company simply prepared the schedule to install 10 meters each day
- Pipeco will spend $1,000 each day

$100,000	= Budget	→ BAC
$30,000	= Planned Value	→ PV
$20,000	= Actual Cost	→ AC
$40,000	= Earned Value	→ EV

Example: Pipeco Incorporated has been awarded a piping project to install 1,000 meters of pipeline in 100 days with a cost of $100,000. Today is the 30th day of the project, and Pipeco has installed 400 meters of piping work up to now. Also, the report says $20,000 has been spent until now. Let's do the EVM calculations.

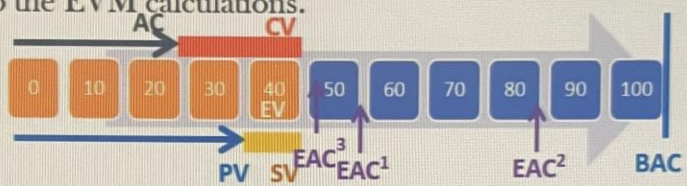


BAC = $100,000; PV = $30,000
AC = $20,000; EV = $40,000

1) Cost Variance (CV): CV = EV - AC
= $40,000 - $20,000
= $20,000
CV > 0 → GOOD!

Tools and Techniques

Example: Pipeco Incorporated has been awarded a piping project to install 1,000 meters of pipeline in 100 days with a cost of $100,000. Today is the 30th day of the project, and Pipeco has installed 400 meters of piping work up to now. Also, the report says $20,000 has been spent until now. Let's do the EVM calculations.



BAC = $100,000; PV = $30,000; CPI = 2
AC = $20,000; EV = $40,000; SPI = 1.33

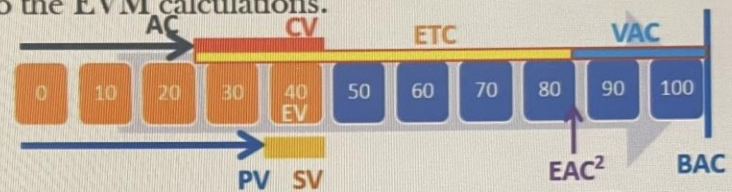
Assume we have re-estimated the future cost from now on as $50,000, calculate the estimated total cost at the end of the project.

Estimate at Completion (EAC)

EAC = AC + New Cost Estimation

= $20,000 + $50,000
= **$70,000**

Example: Pipeco Incorporated has been awarded a piping project to install 1,000 meters of pipeline in 100 days with a cost of $100,000. Today is the 30th day of the project, and Pipeco has installed 400 meters of piping work up to now. Also, the report says $20,000 has been spent until now. Let's do the EVM calculations.



BAC = $100,000; PV = $30,000; CPI = 2
AC = $20,000; EV = $40,000; SPI = 1.33

If EAC=$80,000, how much under or over budget will the estimation be when we complete the Project?

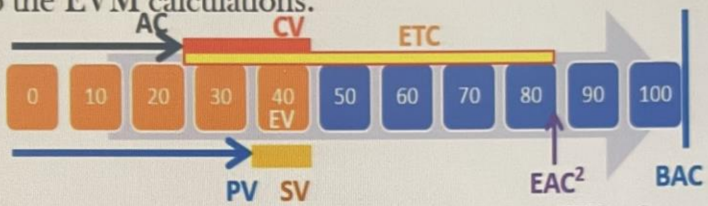
Variance at Completion (VAC)

VAC = BAC - EAC

= $100,000 - $80,000
= **$20,000**

Tools and Techniques

Example: Pipeco Incorporated has been awarded a piping project to install 1,000 meters of pipeline in 100 days with a cost of \$100,000. Today is the 30th day of the project, and Pipeco has installed 400 meters of piping work up to now. Also, the report says \$20,000 has been spent until now. Let's do the EVM calculations.



BAC = \$100,000; PV = \$30,000; CPI = 2
AC = \$20,000; EV = \$40,000; SPI = 1.33

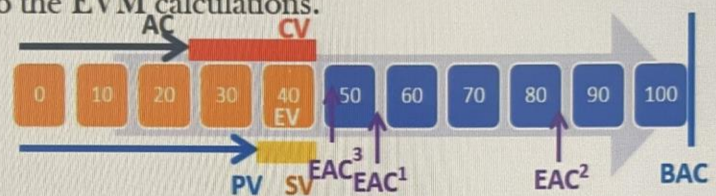
If EAC=\$80,000, how much more should we spend to complete the project?

Estimate to Complete (ETC)

ETC = EAC – AC

= \$80,000 - \$20,000
= **\$60,000**

Example: Pipeco Incorporated has been awarded a piping project to install 1,000 meters of pipeline in 100 days with a cost of \$100,000. Today is the 30th day of the project, and Pipeco has installed 400 meters of piping work up to now. Also, the report says \$20,000 has been spent until now. Let's do the EVM calculations.



BAC = \$100,000; PV = \$30,000; CPI = 2
AC = \$20,000; EV = \$40,000; SPI = 1.33

If we want to complete the project with a cost equal to the budget, what should be the remaining CPI?

To Complete Performance Index (TCPI)

TCPI = (BAC – EV) / (BAC – AC)

= (\$100,000 - \$40,000) / (\$100,000 - \$20,000)
= \$60,000 / \$80,000
= **0.75**